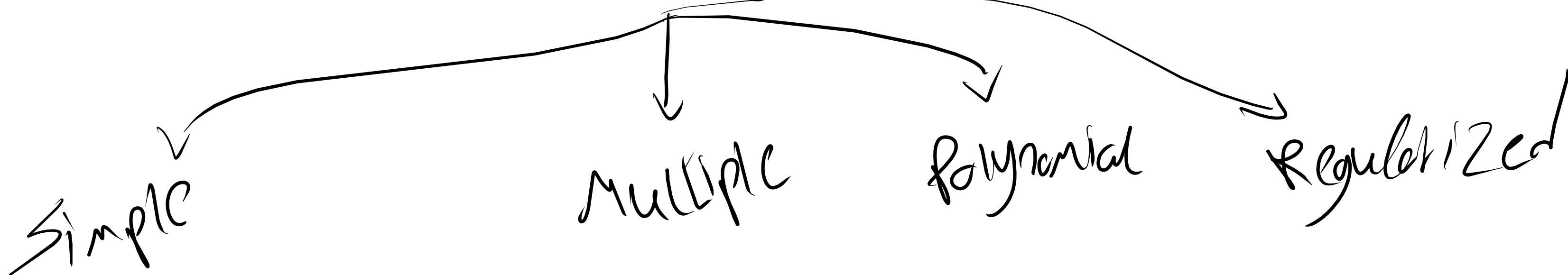
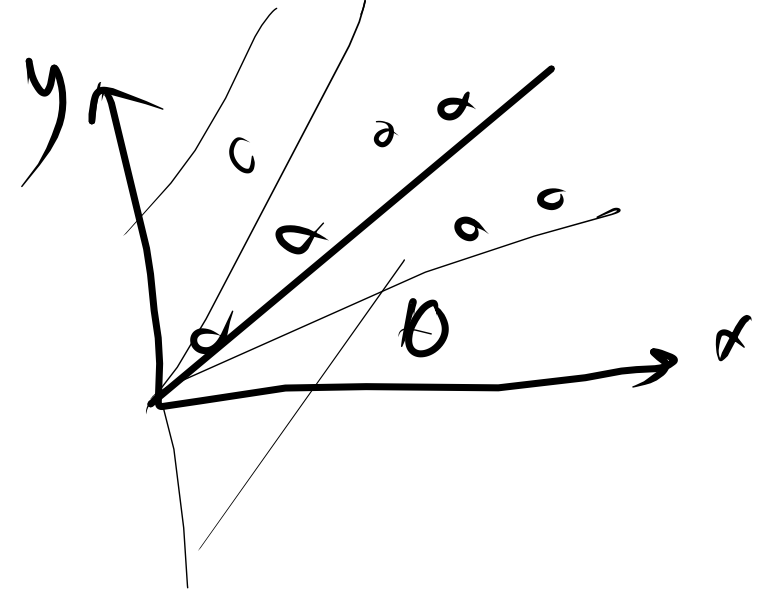


x, y

Supervised Regression

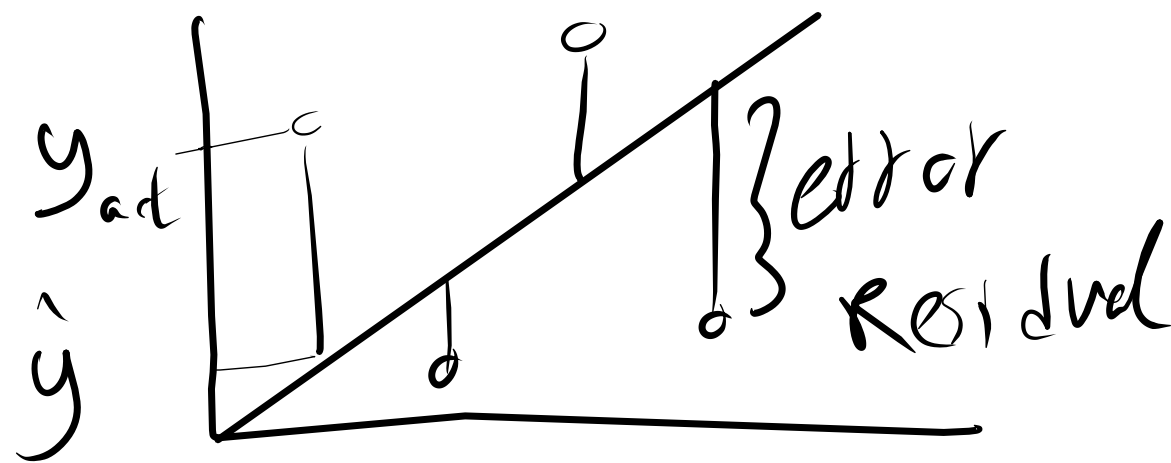
Linear Regression





$$\hat{y} = \theta_0 + \theta_1 x$$

intercept slope



$$RSS = \sum_{i=1}^n (\hat{y} - y_i)^2 = \sum_{i=1}^n [(\theta_0 + \theta_1 x) - y_i]^2$$

Solve for $\theta_0, \theta_1 \Rightarrow RSS \downarrow \downarrow$

Solve for $\theta_0, \theta_1 \Rightarrow R_{ss} \downarrow$

Analytical solution

OLS

$$\theta_1 = \frac{\sum_{i=1}^n (\bar{x} - x_i)(\bar{y} - y_i)}{\sum_{i=1}^n (\bar{x} - x_i)^2}$$

$$\theta_0 = \bar{y} - \theta_1 \bar{x}$$

optimization sol
Gradient Descent

$$J_{\theta_0, \theta_1} = \frac{1}{2n} \sum_{i=1}^n [(\theta_0 + \theta_1 x_i) - y_i]^2$$

1- Initiate random values for theta_0 and theta_1

2- Calculate Loss

3- Update theta_0 and theta_1

$$\theta_{0_{new}} = \theta_{0_{old}} - \alpha \frac{\partial J}{\partial \theta_0} = \theta_{0_{old}} - \alpha \frac{1}{2n} \sum_{i=1}^n [(\theta_0 + \theta_1 x_i) - y_i]$$

$$\theta_{1_{new}} = \theta_{1_{old}} - \alpha \frac{1}{2n} \sum_{i=1}^n [(\theta_0 + \theta_1 x_i) - y_i] x_i$$

Multiple Linear Regression

$$y = \theta_0 + \underbrace{\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n}_{\text{weights}}$$

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$

$$\theta = [\theta_0, \theta_1, \theta_2, \theta_3, \dots, \theta_n]$$

$$X = \begin{bmatrix} x_1 & x_2 & x_3 & \dots & x_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_1 & x_2 & x_3 & \dots & x_n \end{bmatrix}$$

$$\hat{y} = \theta^T X$$

Array

1-D (Vector)

$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{3,1}$

2-D (Matrix)

$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2,3}$

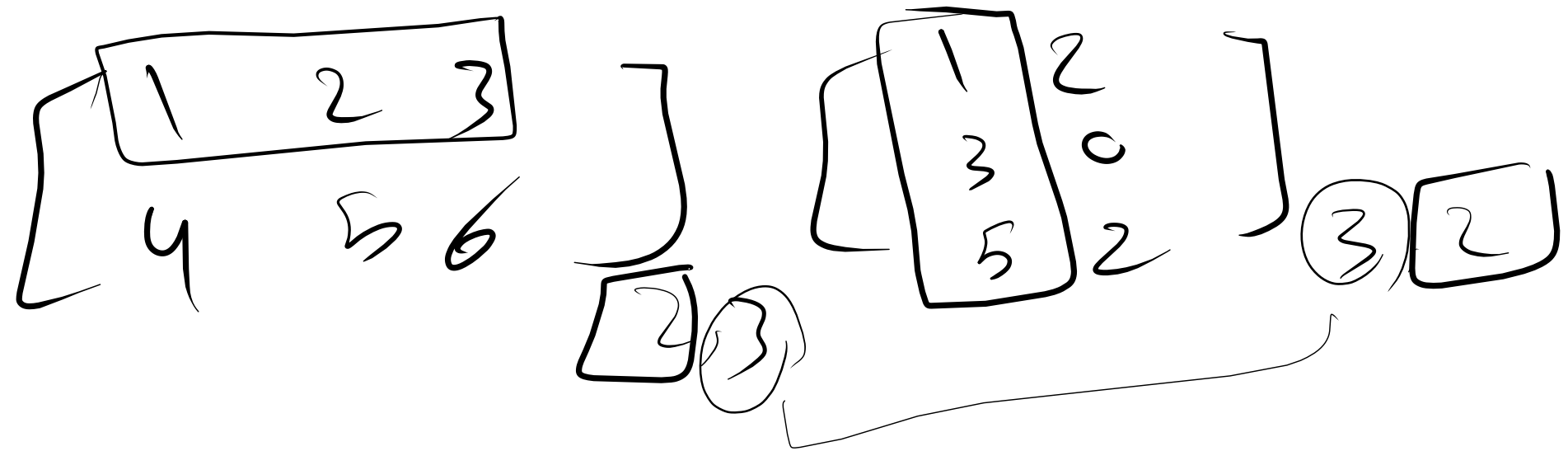
3-D

... n-D

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot 10 = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} = 4 + 2 + 0 = 6$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot 10 = \begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \end{bmatrix}$$



Rule : Num of cols of first matrix = Num of rows of second matrix

Shape : Num of rows of first matrix and Num of cols of second matrix



$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2,3}, \quad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}_{3,2}$$

$$\hat{y} = \theta^T X$$

$$\theta = (X^T X)^{-1} \cdot X^T \cdot y$$

Gradient Descent

$$\theta_{new} = \theta_{old} - \frac{\alpha}{2n} \sum_{i=1}^n [(\theta_0 + \theta_1 x_1 + \theta_2 x_2) - y_i] x_i^M$$

- 1- Initiate random values for theta_0 and theta_1
- 2- Calculate Loss
- 3- Update theta_0 and theta_1

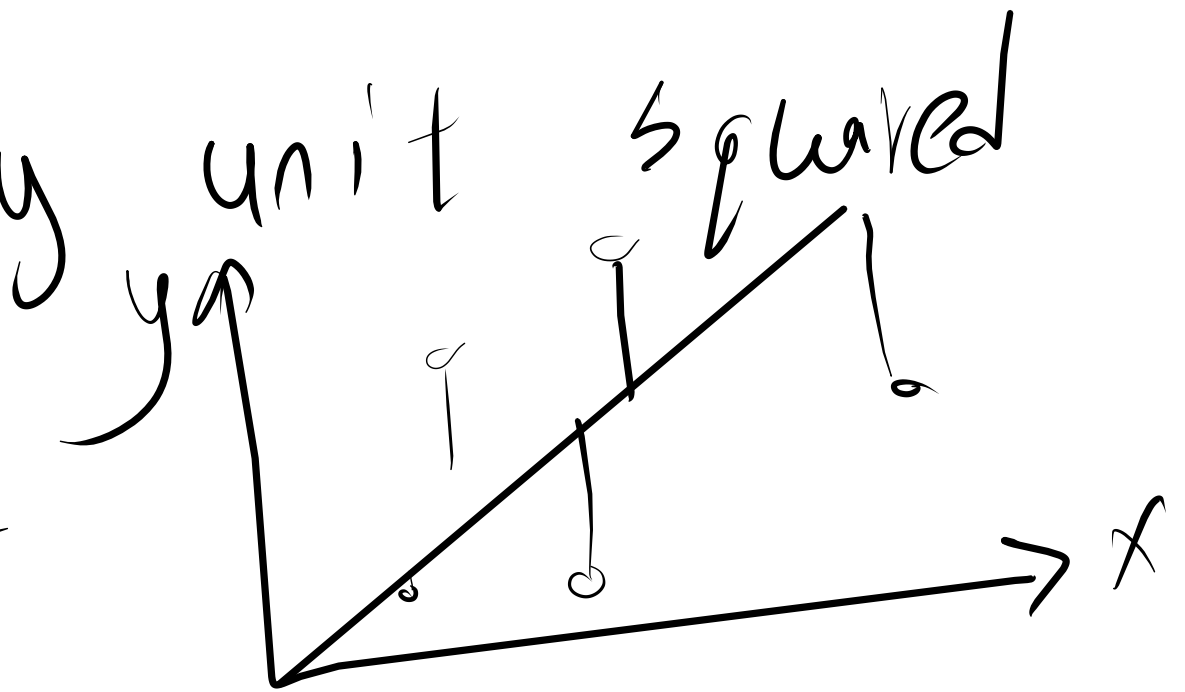
Gradient Descent

Batch

Mini Batch

Stochastic

Regression Evaluation Metrics

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y} - y_i)^2 \rightarrow y \text{ unit squared}$$
$$RMSE = \frac{1}{n} \sum_{i=1}^n \sqrt{(\hat{y} - y_i)^2}$$


R2

$$R^2 = 1 - \frac{\sum \sum E}{\sum \sum T}$$

$(-\infty \rightarrow 1)$

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (\bar{y} - y_i)^2}$$

